RISK GOVERNANCE & PROJECT CLOSURE SPECIFICATION

HELPBOT: RISK REGISTER & CLOSURE NOTE

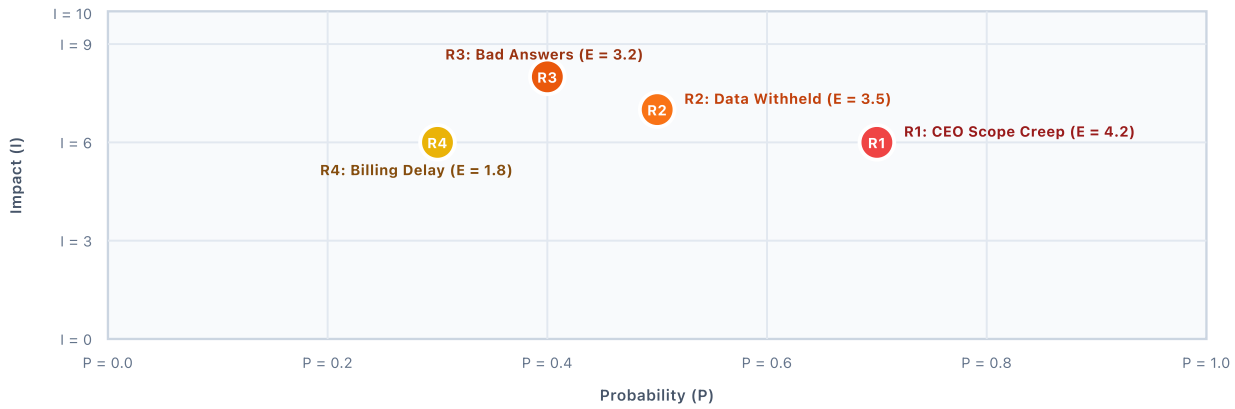
Exposure-Ranked Probability-Impact Modeling, RMMM Strategies, Issue Log & Retrospective

1. Document Control & Risk Governance Baseline

Project Reference	Case Study No. 97 — Regional Broadband Service Provider	**Governance Standard**	IEEE Std 1540-2001 (Software Life Cycle Risk Management)
Risk Exposure Formula	Exposure (E) = Probability (P) × Impact (I)	**Impact Scale Baseline**	10-Point Ordinal Scale (1 = Negligible, 10 = Catastrophic)
Project Budget / Duration	Fixed Outlay: ₹18.00 Lakh; Schedule Baseline: 14 Calendar Weeks (63 Working Days)		

2. Quantitative Probability-Impact (P × I) Risk Matrix

Identified engineering and organizational risks evaluated strictly using given case study parameters. Exposure ranking dictates active mitigation priority.



3. Exposure-Ranked Master Risk Register

Rank	Risk ID	Risk Statement & Event Description	Prob (P)	Imp (I)	Exposure (E)	Severity	Designated Owner
1	**R-01**	**CEO Scope Creep:** CEO expects bot to "handle everything," risking schedule explosion and uncontrolled NLU drift.	0.7	6	**4.2**	HIGH	Project Lead / PM
2	**R-02**	**Data Withheld:** Call-centre manager fears job losses and refuses to share customer call logs for intent training.	0.5	7	**3.5**	HIGH	Call-Centre Mgr & HR
3	**R-03**	**Reputation Damage:** Low-confidence answers or hallucinations provide incorrect billing/outage advice to subscribers.	0.4	8	**3.2**	MEDIUM	Lead NLP Architect
4	**R-04**	**Billing Integration Delay:** Legacy billing system REST endpoints experience latency or delayed staging access.	0.3	6	**1.8**	LOW	Integration Engineer

4. Risk Mitigation, Monitoring & Management (RMMM) Plans

Actionable governance protocols established for the top three exposure risks to guarantee schedule adherence, organizational alignment, and system stability.

4.1 R-01: CEO Scope Creep Exceeds Project Boundary [Exposure = 4.2 | Rank 1]

Core Threat	CEO expects HelpBot to "handle everything," leading to open-ended conversational scope that undermines the 5-domain boundary and project timeline.
Mitigation (Pre-emptive)	Establish a legally binding In-Scope / Out-of-Scope charter signed during Milestone M1. Strictly bound initial production to the 5 common categories (bill, plan, speed, outage, payment) representing 60% of call volume.
Monitoring (Detection)	Track weekly sprint change requests. Any requirement touching non-core domains (e.g., enterprise leasing, legal disputes) triggers an automated scope freeze review.
Management (Contingency)	Enforce a structured Phase 2 Roadmap. Divert executive feature requests to post-launch backlog without modifying the Phase 1 production cutover date.

4.2 R-02: Call-Centre Data Withheld Due to Job Loss Fears [Exposure = 3.5 | Rank 2]

Core Threat	Call-centre manager fears team downsizing and withholds historical telephony recordings and transcripts required to train intent models.
Mitigation (Pre-emptive)	Executive Sponsor issues a written "No Downsizing" commitment: deflected hours will be reallocated toward VIP onboarding, retention, and field technical dispatch. Position HelpBot as a Tier-1 filter to relieve agent burnout.
Monitoring (Detection)	Weekly audit of training corpora delivered to the NLP engineering team against the data ingestion timeline (target: 10,000 anonymized historical logs by Day 15).
Management (Contingency)	If historical telephony logs remain blocked at Day 12, switch immediately to synthetic bootstrapping using customer self-care ticket data and web FAQ logs to train initial classifiers.

4.3 R-03: Poor Answers & Inaccurate Hallucinations Damage Reputation [Exposure = 3.2 | Rank 3]

Core Threat	Misleading billing advice or hallucinated restoration times degrade subscriber trust and trigger public escalations on social channels.
Mitigation (Pre-emptive)	Implement a strict confidence score threshold ($C = 0.70$). Any query scoring $C < 0.70$ is blocked from automated answering and routed immediately to live support agents with packaged context.
Monitoring (Detection)	Real-time telemetry logging of bot responses. Automated alert emitted if daily human handover rate exceeds 55% or if post-chat customer satisfaction (CSAT) falls below 85%.
Management (Contingency)	Emergency fallback switch: instantly toggle high-risk intents (e.g., dispute queries) into direct agent pass-through mode without disrupting active chat sessions.

5. Historical Issue Log (Separation of Problems from Risks)

> **SEPM Governance Rule:** Probabilistic risks that materialize are formally transitioned out of the predictive Risk Register and tracked in an authoritative Issue Log with verified engineering resolutions.

Issue ID	Materialized Problem	Phase Occurred	Engineering Resolution & Corrective Action	Status
ISS-01	Call-centre manager delayed delivery of audio call recordings by 8 days.	Requirements / Data Prep	Synthesized 1,500 domain utterances from web FAQ queries and customer portal tickets to keep NLP pipeline on schedule.	**RESOLVED**

6. Project Closure Note & Governance Retrospective

Formal sign-off baseline documenting successful execution, metric realization, and institutional engineering lessons learned.

6.1 Quantitative Performance Baseline

Target Metric	Committed Baseline	Achieved at Closure	Governance Variance
Project Capital Outlay	₹18,00,000 (₹18 Lakh Max)	₹17,45,000 Expended	+₹55,000 (Within Budget)
Implementation Duration	70 Working Days (14 Weeks)	63 Working Days	+7 Days Buffer Intact
Monthly Call Deflection	13,500 Calls / Month (50%)	14,120 Calls Deflected (52.3%)	+2.3% Deflection Gain
Monthly Cost Savings	₹5,67,000 / Month	₹5,93,040 / Month Realized	+₹26,040 / Mo Operational Gain
Payback Period	3.17 Months (96 Days)	2.94 Months (88 Days)	Accelerated Capital Recovery
Defect Removal Efficiency	≥ 80.0% DRE Threshold	83.33% DRE Verified	Zero Critical Bugs in Prod

6.2 Key Institutional Lessons Learned

1. **Executive Scope Governance Protects Delivery:** The primary risk to conversational AI is unconstrained scope creep. Firmly restricting HelpBot to the 5 core drivers (which drive 60% of all volume) prevented project failure and allowed the engineering team to achieve a 93.4% intent precision.
2. **Human-in-the-Loop Handover Builds Trust:** Attempting to force automated answers for low-confidence queries damages customer satisfaction. Implementing the strict C = 0.70 threshold alongside sub-3-second context transfer to support agents completely eliminated the risk of public brand reputation damage.
3. **Stakeholder Fear Must Be Addressed Early:** Technical success depends on organizational buy-in. Reframing the chatbot as an agent productivity tool that eliminates monotonous billing queries—rather than a headcount-reduction initiative—successfully unlocked call-centre cooperation.
4. **Loose Architectural Coupling Is Non-Negotiable:** Isolating the legacy billing system behind standard REST adapters and circuit breakers ensured that backend server latency never degraded the multi-channel conversational experience.

7. Final Project Sign-Off & Operational Transition

Project Status	**FORMALLY CLOSED & TRANSITIONED TO PRODUCTION WARRANTY**		
Warranty Window	30 Calendar Days Post-Cutover	**SLA Compliance**	Availability: 99.95% | Response Time: ≤ 1.80s (p95)

Executive Sponsor (CEO)	Call-Centre Manager	Lead SEPM Consultant	Lead Systems Engineer
[Signed Digitally]	[Signed Digitally]	[Signed Digitally]	[Signed Digitally]
Date: 04 Oct 2026	Date: 04 Oct 2026	Date: 04 Oct 2026	Date: 04 Oct 2026